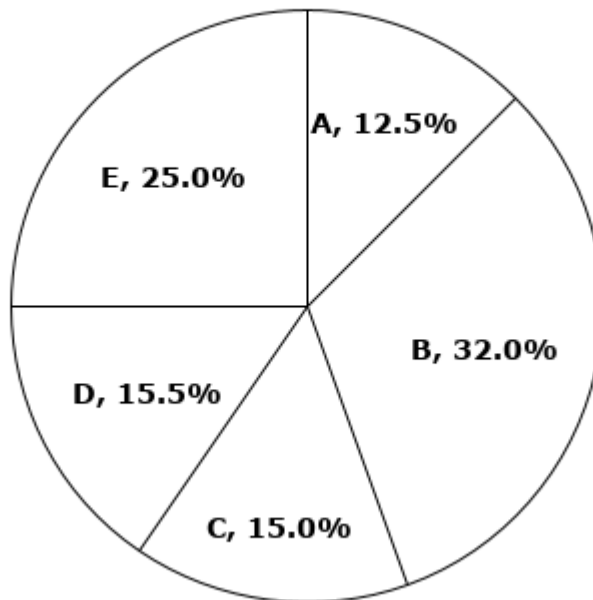


1. Questions

Study the following information carefully and answer the questions given below.

The given pie chart shows the percentage of plastic bottles sold by five different companies A, B, C, D and E.

The percentage of plastic bottles sold



Note: The difference between the number of plastic bottles sold by companies B and E is 140.

If the ratio of the number of plastic bottles to steel bottles sold by company B is 4 : 3. Find the difference between the number of plastic bottles sold by company A and the number of steel bottles sold by company B.

- a. 130
- b. 250
- c. 150
- d. 230
- e. 180

@RankZen Banker

2. Questions

Find the ratio between the number of plastic bottles sold by companies A and C.

- a. 5 : 6
- b. 4 : 3
- c. 6 : 7
- d. 2 : 3

e. 5 : 2

3. Questions

The number of plastic bottles sold by company D is what percentage more or less than the number of plastic bottles sold by company C?

- a. 3.33% less
- b. 2.89% less
- c. 3.33% more
- d. 1.47% more
- e. 1.25% more

4. Questions

What is the difference between the total number of plastic bottles sold by companies B and C together and the total number of plastic bottles sold by companies A and D together?

- a. 250
- b. 180
- c. 320
- d. 220
- e. 380

5. Questions

If the average number of plastic bottles sold by companies C and E together is 2y, then find the value of y.

- a. 100
- b. 200
- c. 300
- d. 500
- e. 400

@RankZen Banker

6. Questions

Study the following information carefully and answer the questions given below.

The table shows the total number of students (boys + girls) in five different schools, namely A, B, C, D, and E in a city and the ratio of boys to girls in each school.

School	The total number of students in each school	The ratio of the number of boys to the number of girls
A	720	5 : 4
B	640	5 : 3
C	900	2 : 3
D	600	7 : 5
E	840	4 : 3

Note: The total number of students in a school = The number of boys + The number of girls.

What is the total number of girls in School B and School C together?

- a. 580
- b. 820
- c. 640
- d. 780
- e. 690

7. Questions

The number of boys in School A is what percentage of the number of boys in School E?

- a. 83.33%
- b. 60%
- c. 55.55%
- d. 67.5%
- e. 80%

@RankZen Banker

8. Questions

What is the average number of boys in schools B, C and D together?

- a. 360
- b. 320
- c. 370

d. 340

e. 390

9. Questions

What is the ratio of the number of girls in School D to the number of girls in School E?

a. 36 : 17

b. 14 : 5

c. 17 : 9

d. 15 : 14

e. 25 : 36

10. Questions

If 20% of the total students in School C leave the school and join School B, find the new total number of students in School B.

a. 650

b. 820

c. 780

d. 890

e. 710

11. Questions

R and S started a business with investments of Rs. 2000 and Rs. 9000, respectively. After y months S left the business. If at the end of one year, the profit share of R is Rs 1600 out of a total profit of Rs 4600, then find the value of y.

a. 9

b. 4

c. 8

d. 5

e. 7

@RankZen Banker

12. Questions

Jay invested Rs. X in scheme A at 12% p.a. compound interest (compounded annually) for 2 years. If he received an interest of Rs. 3180, find the value of X.

a. Rs. 15500

b. Rs. 11200

- c. Rs. 10000
- d. Rs. 12500
- e. Rs. 10800

13. Questions

An article is sold at a 15% profit. If it had been sold for Rs. 27 more, the profit percent would have been 24%. Find the cost price of the article.

- a. Rs. 270
- b. Rs. 510
- c. Rs. 300
- d. Rs. 480
- e. Rs. 350

14. Questions

A car covers 280 km in 4 hours and the speed of a bike is $\frac{4}{5}$ of the speed of the car. Find the speed of the bike.

- a. 48 km/hr
- b. 56 km/hr
- c. 32 km/hr
- d. 72 km/hr
- e. 64 km/hr

15. Questions

A 150 liters mixture of juice and water contains 20% water. 30 liters of the mixture are removed, and 'x' liters of water are added to the remaining mixture. If in the resultant mixture water becomes 60%, then find the value of x.

- a. 120
- b. 80
- c. 140
- d. 60
- e. 150

@RankZen Banker

16. Questions

The side of a square is equal to the breadth of a rectangle and the perimeter of the square is 56 cm. If the area of the rectangle is 238 cm^2 , then find the length of the rectangle.

- a. 14 cm
- b. 18 cm
- c. 16 cm
- d. 15 cm
- e. 17 cm

17. Questions

The average of four consecutive odd numbers is 26, and the average of six consecutive even numbers is 31. Find the sum of the smallest odd number and the largest even number.

- a. 53
- b. 37
- c. 45
- d. 59
- e. 47

18. Questions

Three pipes X, Y and Z can fill a tank alone in 4 hours, 6 hours and 12 hours respectively. Find the time taken by all three pipes to fill the tank together.

- a. 5 hours
- b. 3 hours
- c. 2 hours
- d. 4 hours
- e. 7 hours

19. Questions

The ratio of P to (P + 40) is 2 : 3 respectively and the ratio of (P + 25) and (Q + 5) is 3 : 2 respectively. Find the sum of P and Q?

@RankZen Banker

- a. 120
- b. 145
- c. 130
- d. 125
- e. 140

20. Questions

The ratio of the present ages of Rohan and Sohan is 3 : 5 respectively. Mohan is 10 years older than Rohan and 4 years younger than Sohan. Find the age of Sohan five years ago.

- a. 30 years
- b. 28 years
- c. 21 years
- d. 32 years
- e. 35 years

21. Questions

What value should come in the place of (?) in the following questions?

$$(13/4) * (16/5) * (25/13) * 40 = ?$$

- a. 500
- b. 700
- c. 600
- d. 800
- e. 300

22. Questions

$$\sqrt{196} + 480 \div 24 + 55 = 41 + ?$$

- a. 39
- b. 48
- c. 42
- d. 35
- e. 51

23. Questions

$$8^3 - 45 - 24\% \text{ of } 800 = ? - [3/4] * 60$$

- a. 320
- b. 180
- c. 240
- d. 380
- e. 420

@RankZen Banker

24. Questions

$$(528 / 12) \div (11 / 22) = ? - 35\% \text{ of } 800$$

- a. 312
- b. 294
- c. 368
- d. 264
- e. 398

25. Questions

$$1470 \div 6 + \sqrt{4225} - \sqrt{2304} = ?$$

- a. 258
- b. 214
- c. 218
- d. 294
- e. 262

26. Questions

What approximate value will come in the place of (?) in the following questions?

$$(14.99 * 19.98) \div 29.97 = ?^2 - 89.92$$

- a. 20
- b. 18
- c. 15
- d. 10
- e. 30

27. Questions

$$749.93 \div 24.99 + 14.98^2 = ?$$

- a. 255
- b. 319
- c. 261
- d. 325
- e. 218

@RankZen Banker

28. Questions

$$17.98 * 14.99 + ?\% \text{ of } 399.92 = 469.96$$

- a. 40
- b. 30
- c. 50
- d. 20
- e. 60

29. Questions

$$\sqrt{1023.92} + \sqrt{483.99} - \sqrt{120.96} = ?$$

- a. 53
- b. 39
- c. 61
- d. 28
- e. 43

30. Questions

$$419.98 \div 6.97 * 11.98 = ? - 149.91$$

- a. 910
- b. 870
- c. 730
- d. 820
- e. 970

31. Questions

What value will come in the place of (?) in the following number series?

$$17, 19, 22, 27, ?, 45$$

- a. 29
- b. 32
- c. 28
- d. 34
- e. 27

32. Questions**4, 7, 13, ?, 49, 97**

- a. 25
- b. 39
- c. 28
- d. 35
- e. 42

33. Questions**9, ?, 14, 23, 39, 64**

- a. 9
- b. 12
- c. 10
- d. 8
- e. 15

34. Questions**19, 25, 37, 55, 79, ?**

- a. 115
- b. 104
- c. 111
- d. 120
- e. 109

35. Questions**12, 22, 42, ?, 112, 162**

- a. 58
- b. 72
- c. 62
- d. 47
- e. 69

36. Questions

@RankZen Banker

Find out the wrong number in the following number series.

34, 23, 45, 12, 56, 2

- a. 2
- b. 34
- c. 12
- d. 23
- e. 56

37. Questions

14, 7, 8, 12, 14, 35, 39

- a. 12
- b. 8
- c. 7
- d. 39
- e. 35

38. Questions

43, 60, 80, 102, 131, 162

- a. 43
- b. 80
- c. 131
- d. 60
- e. 162

39. Questions

3, 9, 23, 68, 203, 608

- a. 3
- b. 68
- c. 203
- d. 23
- e. 9

40. Questions

@RankZen Banker

1, 2, 10, 48, 384, 3840

- a. 2
- b. 384
- c. 10
- d. 48
- e. 1

41. Questions

The following questions contain two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

I). $x^2 - 13x + 42 = 0$

II). $y^2 - 11y + 28 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or a relationship cannot be established
- d. $x < y$
- e. $x \leq y$

42. Questions

I). $x^2 - 27x + 180 = 0$

II). $y^2 - 17y + 70 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or a relationship cannot be established
- d. $x < y$
- e. $x \leq y$

@RankZen Banker

43. Questions

I). $2x^2 + 5x + 3 = 0$

II). $y^2 + 4y + 4 = 0$

- a. $x < y$
- b. $x \geq y$

- c. $x = y$ or a relationship cannot be established
- d. $x > y$
- e. $x \leq y$

44. Questions

I). $x^2 - 3x - 10 = 0$

II). $y^2 + 3y + 2 = 0$

- a. $x < y$
- b. $x \geq y$
- c. $x = y$ or a relationship cannot be established
- d. $x > y$
- e. $x \leq y$

45. Questions

I). $x^2 - 16x + 63 = 0$

II). $y^2 - 21y + 108 = 0$

- a. $x < y$
- b. $x \geq y$
- c. $x = y$ or a relationship cannot be established
- d. $x > y$
- e. $x \leq y$

Explanations:

1. Questions

@RankZen Banker

Let the total number of bottles sold by all the companies together = $100a$

The number of plastic bottles sold by company A = $100a * 12.5\% = 12.5a$

The number of plastic bottles sold by company B = $100a * 32\% = 32a$

The number of plastic bottles sold by company C = $100a * 15\% = 15a$

The number of plastic bottles sold by company D = $100a * 15.5\% = 15.5a$

The number of plastic bottles sold by company E = $100a * 25\% = 25a$

According to the question,

$$32a - 25a = 140$$

$$7a = 140$$

$$a = 20$$

The total number of plastic bottles sold by all companies together = $100 * 20 = 2000$

The number of plastic bottles sold by company A = $12.5 * 20 = 250$

The number of plastic bottles sold by company B = $32 * 20 = 640$

The number of plastic bottles sold by company C = $15 * 20 = 300$

The number of plastic bottles sold by company D = $15.5 * 20 = 310$

The number of plastic bottles sold by company E = $25 * 20 = 500$

Company	The number of plastic bottles sold
A	250
B	640
C	300
D	310
E	500
Total	2000

Answer: D

The number of steel bottles sold by company B = $640 * 3/4 = 480$

The number of plastic bottles sold by company A = 250

The required difference = $480 - 250 = 230$

Hence, the correct answer is option D.

2. Questions

Let the total number of bottles sold by all the companies together = $100a$

The number of plastic bottles sold by company A = $100a * 12.5\% = 12.5a$

The number of plastic bottles sold by company B = $100a * 32\% = 32a$

The number of plastic bottles sold by company C = $100a * 15\% = 15a$

The number of plastic bottles sold by company D = $100a * 15.5\% = 15.5a$

@RankZen Banker

The number of plastic bottles sold by company E = $100a * 25\% = 25a$

According to the question,

$$32a - 25a = 140$$

$$7a = 140$$

$$a = 20$$

The total number of plastic bottles sold by all companies together = $100 * 20 = 2000$

The number of plastic bottles sold by company A = $12.5 * 20 = 250$

The number of plastic bottles sold by company B = $32 * 20 = 640$

The number of plastic bottles sold by company C = $15 * 20 = 300$

The number of plastic bottles sold by company D = $15.5 * 20 = 310$

The number of plastic bottles sold by company E = $25 * 20 = 500$

Company	The number of plastic bottles sold
A	250
B	640
C	300
D	310
E	500
Total	2000

Answer: A

The number of plastic bottles sold by company A = 250

The number of plastic bottles sold by company C = 300

The required ratio = $250 : 300 = 5 : 6$

Hence, the correct answer is option A.

3. Questions

Let the total number of bottles sold by all the companies together = $100a$

The number of plastic bottles sold by company A = $100a * 12.5\% = 12.5a$

The number of plastic bottles sold by company B = $100a * 32\% = 32a$

@RankZen Banker

The number of plastic bottles sold by company C = $100a * 15\% = 15a$

The number of plastic bottles sold by company D = $100a * 15.5\% = 15.5a$

The number of plastic bottles sold by company E = $100a * 25\% = 25a$

According to the question,

$$32a - 25a = 140$$

$$7a = 140$$

$$a = 20$$

The total number of plastic bottles sold by all companies together = $100 * 20 = 2000$

The number of plastic bottles sold by company A = $12.5 * 20 = 250$

The number of plastic bottles sold by company B = $32 * 20 = 640$

The number of plastic bottles sold by company C = $15 * 20 = 300$

The number of plastic bottles sold by company D = $15.5 * 20 = 310$

The number of plastic bottles sold by company E = $25 * 20 = 500$

Company	The number of plastic bottles sold
A	250
B	640
C	300
D	310
E	500
Total	2000

@RankZen Banker

Answer: C

The number of plastic bottles sold by company D = 310

The number of plastic bottles sold by company C = 300

The required percentage = $[(310 - 300) * 100/300] = 3.33\%$ more

Hence, the correct answer is option C.

4. Questions

Let the total number of bottles sold by all the companies together = $100a$

The number of plastic bottles sold by company A = $100a * 12.5\% = 12.5a$

The number of plastic bottles sold by company B = $100a * 32\% = 32a$

The number of plastic bottles sold by company C = $100a * 15\% = 15a$

The number of plastic bottles sold by company D = $100a * 15.5\% = 15.5a$

The number of plastic bottles sold by company E = $100a * 25\% = 25a$

According to the question,

$$32a - 25a = 140$$

$$7a = 140$$

$$a = 20$$

The total number of plastic bottles sold by all companies together = $100 * 20 = 2000$

The number of plastic bottles sold by company A = $12.5 * 20 = 250$

The number of plastic bottles sold by company B = $32 * 20 = 640$

The number of plastic bottles sold by company C = $15 * 20 = 300$

The number of plastic bottles sold by company D = $15.5 * 20 = 310$

The number of plastic bottles sold by company E = $25 * 20 = 500$

Company	The number of plastic bottles sold
A	250
B	640
C	300
D	310
E	500
Total	2000

@RankZen Banker

Answer: E

The total number of plastic bottles sold by companies B and C together = $640 + 300 = 940$

The total number of plastic bottles sold by companies A and D together = $250 + 310 = 560$

The required difference = $940 - 560 = 380$

Hence, the correct answer is option E.

5. Questions

Let the total number of bottles sold by all the companies together = $100a$

The number of plastic bottles sold by company A = $100a * 12.5\% = 12.5a$

The number of plastic bottles sold by company B = $100a * 32\% = 32a$

The number of plastic bottles sold by company C = $100a * 15\% = 15a$

The number of plastic bottles sold by company D = $100a * 15.5\% = 15.5a$

The number of plastic bottles sold by company E = $100a * 25\% = 25a$

According to the question,

$$32a - 25a = 140$$

$$7a = 140$$

$$a = 20$$

The total number of plastic bottles sold by all companies together = $100 * 20 = 2000$

The number of plastic bottles sold by company A = $12.5 * 20 = 250$

The number of plastic bottles sold by company B = $32 * 20 = 640$

The number of plastic bottles sold by company C = $15 * 20 = 300$

The number of plastic bottles sold by company D = $15.5 * 20 = 310$

The number of plastic bottles sold by company E = $25 * 20 = 500$

Company	The number of plastic bottles sold
A	250
B	640
C	300
D	310
E	500
Total	2000

@RankZen Banker

Answer: B

The total number of plastic bottles sold by companies C and E together = $300 + 500 = 800$

According to the question,

$$(800/2) = 2y$$

$$y = 200$$

Hence, the correct answer is option B.

6. Questions

School A:

The total number of students in school A = 720

The number of boys in school A = $720 * \frac{5}{9} = 400$

The number of girls in school A = $720 - 400 = 320$

School B:

The total number of students in school B = 640

The number of boys in school B = $640 * \frac{5}{8} = 400$

The number of girls in school B = $640 - 400 = 240$

School C:

The total number of students in school C = 900

The number of boys in school C = $900 * \frac{2}{5} = 360$

The number of girls in school C = $900 - 360 = 540$

School D:

The total number of students in school D = 600

The number of boys in school D = $600 * \frac{7}{12} = 350$

The number of girls in school D = $600 - 350 = 250$

School E:

The total number of students in school E = 840

The number of boys in school E = $840 * \frac{4}{7} = 480$

The number of girls in school E = $840 - 480 = 360$

@RankZen Banker

School	The number of boys in each school	The number of girls in each school	The total number of students in each school
A	400	320	720
B	400	240	640
C	360	540	900
D	350	250	600
E	480	360	840
Total	1990	1710	3700

Answer: D

The total number of girls in schools B and C together = $240 + 540 = 780$

Hence, the correct answer is option D.

7. Questions

School A:

The total number of students in school A = 720

The number of boys in school A = $720 \times \frac{5}{9} = 400$

The number of girls in school A = $720 - 400 = 320$

School B:

The total number of students in school B = 640

The number of boys in school B = $640 \times \frac{5}{8} = 400$

The number of girls in school B = $640 - 400 = 240$

School C:

The total number of students in school C = 900

The number of boys in school C = $900 \times \frac{2}{5} = 360$

The number of girls in school C = $900 - 360 = 540$

School D:

The total number of students in school D = 600

The number of boys in school D = $600 \times \frac{7}{12} = 350$

@RankZen Banker

The number of girls in school D = $600 - 350 = 250$

School E:

The total number of students in school E = 840

The number of boys in school E = $840 \times \frac{4}{7} = 480$

The number of girls in school E = $840 - 480 = 360$

School	The number of boys in each school	The number of girls in each school	The total number of students in each school
A	400	320	720
B	400	240	640
C	360	540	900
D	350	250	600
E	480	360	840
Total	1990	1710	3700

Answer: A

The number of boys in school A = 400

The number of boys in school E = 480

The required percentage = $400 \times \frac{100}{480} = 83.33\%$

Hence, the correct answer is option A.

8. Questions

School A:

The total number of students in school A = 720

The number of boys in school A = $720 \times \frac{5}{9} = 400$

The number of girls in school A = $720 - 400 = 320$

School B:

The total number of students in school B = 640

The number of boys in school B = $640 \times \frac{5}{8} = 400$

The number of girls in school B = $640 - 400 = 240$

@RankZen Banker

School C:

The total number of students in school C = 900

The number of boys in school C = $900 \times \frac{2}{5} = 360$

The number of girls in school C = $900 - 360 = 540$

School D:

The total number of students in school D = 600

The number of boys in school D = $600 \times \frac{7}{12} = 350$

The number of girls in school D = $600 - 350 = 250$

School E:

The total number of students in school E = 840

The number of boys in school E = $840 \times \frac{4}{7} = 480$

The number of girls in school E = $840 - 480 = 360$

School	The number of boys in each school	The number of girls in each school	The total number of students in each school
A	400	320	720
B	400	240	640
C	360	540	900
D	350	250	600
E	480	360	840
Total	1990	1710	3700

Answer: C

@RankZen Banker

The total number of boys in schools B, C and D together = $400 + 360 + 350 = 1110$

The required average = $1110/3 = 370$

Hence, the correct answer is option C.

9. Questions

School A:

The total number of students in school A = 720

The number of boys in school A = $720 \times \frac{5}{9} = 400$

The number of girls in school A = $720 - 400 = 320$

School B:

The total number of students in school B = 640

The number of boys in school B = $640 \times \frac{5}{8} = 400$

The number of girls in school B = $640 - 400 = 240$

School C:

The total number of students in school C = 900

The number of boys in school C = $900 \times \frac{2}{5} = 360$

The number of girls in school C = $900 - 360 = 540$

School D:

The total number of students in school D = 600

The number of boys in school D = $600 \times \frac{7}{12} = 350$

The number of girls in school D = $600 - 350 = 250$

School E:

The total number of students in school E = 840

The number of boys in school E = $840 \times \frac{4}{7} = 480$

The number of girls in school E = $840 - 480 = 360$

School	The number of boys in each school	The number of girls in each school	The total number of students in each school
A	400	320	720
B	400	240	640
C	360	540	900
D	350	250	600
E	480	360	840
Total	1990	1710	3700

Answer: E

The number of girls in school D = 250

The number of girls in school E = 360

The required ratio = $250 : 360 = 25 : 36$

Hence, the correct answer is option E.

10. Questions

School A:

The total number of students in school A = 720

The number of boys in school A = $720 * \frac{5}{9} = 400$

The number of girls in school A = $720 - 400 = 320$

School B:

The total number of students in school B = 640

The number of boys in school B = $640 * \frac{5}{8} = 400$

The number of girls in school B = $640 - 400 = 240$

School C:

The total number of students in school C = 900

The number of boys in school C = $900 * \frac{2}{5} = 360$

The number of girls in school C = $900 - 360 = 540$

School D:

The total number of students in school D = 600

The number of boys in school D = $600 * \frac{7}{12} = 350$

The number of girls in school D = $600 - 350 = 250$

School E:

The total number of students in school E = 840

The number of boys in school E = $840 * \frac{4}{7} = 480$

The number of girls in school E = $840 - 480 = 360$

@RankZen Banker

School	The number of boys in each school	The number of girls in each school	The total number of students in each school
A	400	320	720
B	400	240	640
C	360	540	900
D	350	250	600
E	480	360	840
Total	1990	1710	3700

Answer: B

The number of new students in school B = $900 * 20\% = 180$

The new total number of students in school B = $640 + 180 = 820$

Hence, the correct answer is option B.

11. Questions

Answer: D

The total investment of R = $2000 * 12 = \text{Rs. } 24000$

The total investment of S = $9000 * y = \text{Rs. } 9000y$

According to the question,

Profit share of R and S

$$24000/9000y = 1600/(4600 - 1600)$$

$$24/9y = 1600/3000$$

$$8/3y = 8/15$$

$$y = 5$$

Hence, the correct answer is option D.

12. Questions

Answer: D

Formula used:

$$\text{The compound interest} = P[(1 + R/100)^n - 1]$$

@RankZen Banker

According to the question,

$$X * [(1 + 12/100)^2 - 1] = 3180$$

$$X * (784 - 625)/625 = 3180$$

$$X * 159/625 = 3180$$

$$X = 12500$$

Hence, the correct answer is option D.

13. Questions

Answer: C

Let the cost price of the article = 100a

The selling price of the article = $100a * 115/100 = 115a$

According to the question,

$$(115a + 27) = 100a * 124/100$$

$$115a + 27 = 124a$$

$$9a = 27$$

$$a = 3$$

The cost price of the article = $100a = 100 * 3 = \text{Rs. } 300$

Hence, the correct answer is option C.

14. Questions

Answer: B

The speed of the car = $280/4 = 70 \text{ km/hr}$

The speed of the bike = $70 * 4/5 = 56 \text{ km/hr}$

Hence, the correct answer is option B.

15. Questions

Answer: A

The quantity of water in the initial mixture = $150 * 20\% = 30 \text{ litres}$

The quantity of juice in the initial mixture = $150 * 80\% = 120 \text{ litres}$

The quantity of juice removed = $30 * 80\% = 24 \text{ litres}$

The quantity of water removed = $30 * 20\% = 6 \text{ litres}$

According to the question,

$$(120 - 24)/(30 - 6 + x) = 40/60$$

$$96/(24 + x) = 2 / 3$$

@RankZen Banker

$$144 = 24 + x$$

$$120 = x$$

Hence, the correct answer is option A.

16. Questions

Answer: E

Formula used:

The perimeter of square = $4a$

The area of the rectangle = length * breadth

According to the question,

$$4a = 56$$

$$a = 14$$

The breadth of the rectangle = 14 cm

The length of the rectangle = $238/14 = 17$ cm

Hence, the correct answer is option E.

17. Questions

Answer: D

Let the four consecutive odd numbers = $a, a + 2, a + 4, a + 6$

The total of all odd numbers = $26 * 4 = 104$

$$a + a + 2 + a + 4 + a + 6 = 104$$

$$4a + 12 = 104$$

$$4a = 92$$

$$a = 23$$

The smallest odd number = 23

Let the six even numbers = $b, b + 2, b + 4, b + 6, b + 8, b + 10$

The total of all even numbers = $31 * 6 = 186$

$$b + b + 2 + b + 4 + b + 6 + b + 8 + b + 10 = 186$$

$$6b + 30 = 186$$

$$6b = 156$$

$$b = 26$$

The largest even number = $b + 10 = 26 + 10 = 36$

The required sum = $23 + 36 = 59$

@RankZen Banker

Hence, the correct answer is option D.

18. Questions

Answer: C

Let the total capacity of the tank = 12 units [LCM of 4, 6 and 12]

The efficiency of pipe X = $12/4 = 3$ units per hour

The efficiency of pipe Y = $12/6 = 2$ units per hour

The efficiency of pipe Z = $12/12 = 1$ unit per hour

The combined efficiency of pipes X, Y and Z = $3 + 2 + 1 = 6$ units per hour

The required time = $12/6 = 2$ hours

Hence, the correct answer is option C.

19. Questions

Answer: B

According to the question,

$$P/(P + 40) = 2 / 3$$

$$3P = 2P + 80$$

$$P = 80$$

And,

$$(80 + 25)/(Q + 5) = 3 / 2$$

$$105 / (Q + 5) = 3 / 2$$

$$Q + 5 = 70$$

$$Q = 65$$

The required sum = $80 + 65 = 145$

Hence, the correct answer is option B.

20. Questions

Answer: A

Let the present age of Rohan = $3a$

The present age of Sohan = $5a$

According to the question,

$$(3a + 10) = (5a - 4)$$

$$2a = 14$$

$$a = 7$$

@RankZen Banker

The present age of Sohan = $5 * 7 = 35$ years

The age of Sohan 5 years ago = $35 - 5 = 30$ years

Hence, the correct answer is option A.

21. Questions

Answer: D

$$(13/4) * (16/5) * (25/13) * 40 = ?$$

$$4 * 5 * 40 = ?$$

$$800 = ?$$

Hence, the correct answer is option D.

22. Questions

Answer: B

$$\sqrt{196} + 480 \div 24 + 55 = 41 + ?$$

$$14 + 20 + 55 = 41 + ?$$

$$89 = 41 + ?$$

$$48 = ?$$

Hence, the correct answer is option B.

23. Questions

Answer: A

$$8^3 - 45 - 24\% \text{ of } 800 = ? - [3/4] * 60$$

$$512 - 45 - 192 = ? - 45$$

$$275 = ? - 45$$

$$320 = ?$$

Hence, the correct answer is option A.

24. Questions

Answer: C

$$(528 / 12) \div (11 / 22) = ? - 35\% \text{ of } 800$$

$$44 \div (1/2) = ? - 280$$

$$88 = ? - 280$$

$$? = 368$$

Hence, the correct answer is option C.

25. Questions

@RankZen Banker

Answer: E

$$1470 \div 6 + \sqrt{4225} - \sqrt{2304} = ?$$

$$245 + 65 - 48 = ?$$

$$262 = ?$$

Hence, the correct answer is option E.

26. Questions

Answer: D

$$(14.99 * 19.98) \div 29.97 = ?^2 - 89.92$$

$$(15 * 20) \div 30 = ?^2 - 90$$

$$300 \div 30 = ?^2 - 90$$

$$10 = ?^2 - 90$$

$$100 = ?^2$$

$$? = 10$$

Hence, the correct answer is option D.

27. Questions

Answer: A

$$749.93 \div 24.99 + 14.98^2 = ?$$

$$750 \div 25 + 15^2 = ?$$

$$30 + 225 = ?$$

$$255 = ?$$

Hence, the correct answer is option A.

28. Questions

Answer: C

$$17.98 * 14.99 + ?\% \text{ of } 399.92 = 469.96$$

$$18 * 15 + ?\% \text{ of } 400 = 470$$

$$270 + ? * 4 = 470$$

$$? * 4 = 200$$

$$? = 50$$

Hence, the correct answer is option C.

@RankZen Banker

29. Questions

Answer: E

$$\sqrt{1023.92} + \sqrt{483.99} - \sqrt{120.96} = ?$$

$$\sqrt{1024} + \sqrt{484} - \sqrt{121} = ?$$

$$32 + 22 - 11 = ?$$

$$32 + 11 = ?$$

$$43 = ?$$

Hence, the correct answer is option E.

30. Questions

Answer: B

$$419.98 \div 6.97 * 11.98 = ? - 149.91$$

$$420 \div 7 * 12 = ? - 150$$

$$60 * 12 = ? - 150$$

$$720 = ? - 150$$

$$? = 870$$

Hence, the correct answer is option B.

31. Questions

Answer: D

$$17 + 2 = 19$$

$$19 + 3 = 22$$

$$22 + 5 = 27$$

$$27 + 7 = 34$$

$$34 + 11 = 45$$

Missing number: 34

Hence, the correct answer is option D.

32. Questions

Answer: A

$$4 * 2 - 1 = 7$$

$$7 * 2 - 1 = 13$$

$$13 * 2 - 1 = 25$$

$$25 * 2 - 1 = 49$$

@RankZen Banker

$$49 * 2 - 1 = 97$$

Or

4	7	13	25	49	97
	3	6	12	24	48
	*2	*2	*2	*2	

Missing number: 25

Hence, the correct answer is option A.

33. Questions

Answer: C

$$9 + 1^2 = 10$$

$$10 + 2^2 = 14$$

$$14 + 3^2 = 23$$

$$23 + 4^2 = 39$$

$$39 + 5^2 = 64$$

Missing number: 10

Hence, the correct answer is option C.

34. Questions

Answer: E

$$19 + 6 = 25$$

$$25 + 12 = 37$$

$$37 + 18 = 55$$

$$55 + 24 = 79$$

$$79 + 30 = 109$$

Missing number: 109

Hence, the correct answer is option E.

35. Questions

Answer: B

$$12 + 10 = 22$$

$$22 + 20 = 42$$

$$42 + 30 = 72$$

@RankZen Banker

$$72 + 40 = 112$$

$$112 + 50 = 162$$

Missing number: 72

Hence, the correct answer is option B.

36. Questions

Answer: A

$$34 - 11 = 23$$

$$23 + 22 = 45$$

$$45 - 33 = 12$$

$$12 + 44 = 56$$

$$56 - 55 = 1$$

Wrong number: 2

Correct number: 1

Hence, the correct answer is option A.

37. Questions

Answer: D

$$14 * 0.5 = 7$$

$$7 + 1 = 8$$

$$8 * 1.5 = 12$$

$$12 + 2 = 14$$

$$14 * 2.5 = 35$$

$$35 + 3 = 38$$

Wrong number: 39

Correct number: 38

Hence, the correct answer is option D.

38. Questions

Answer: B

$$43 + 17 = 60$$

$$60 + 19 = 79$$

$$79 + 23 = 102$$

$$102 + 29 = 131$$

@RankZen Banker

$$131 + 31 = 162$$

Wrong number: 80

Correct number: 79

Hence, the correct answer is option B.

39. Questions

Answer: E

$$3 * 3 - 1 = 8$$

$$8 * 3 - 1 = 23$$

$$23 * 3 - 1 = 68$$

$$68 * 3 - 1 = 203$$

$$203 * 3 - 1 = 608$$

Wrong number: 9

Correct number: 8

Hence, the correct answer is option E.

40. Questions

Answer: C

$$1 * 2 = 2$$

$$2 * 4 = 8$$

$$8 * 6 = 48$$

$$48 * 8 = 384$$

$$384 * 10 = 3840$$

Wrong number: 10

Correct number: 8

Hence, the correct answer is option C.

41. Questions

Answer: C

$$D). x^2 - 13x + 42 = 0$$

$$x^2 - 7x - 6x + 42 = 0$$

$$x(x - 7) - 6(x - 7) = 0$$

$$(x - 6)(x - 7) = 0$$

$$x = 6 \text{ or } 7$$

@RankZen Banker

II). $y^2 - 11y + 28 = 0$

$$y^2 - 7y - 4y + 28 = 0$$

$$y(y - 7) - 4(y - 7) = 0$$

$$(y - 4)(y - 7) = 0$$

$$y = 4 \text{ or } 7$$

So, the relationship cannot be established.

Hence, the correct answer is option C.

42. Questions

Answer: A

I). $x^2 - 27x + 180 = 0$

$$x^2 - 15x - 12x + 180 = 0$$

$$x(x - 15) - 12(x - 15) = 0$$

$$(x - 12)(x - 15) = 0$$

$$x = 12 \text{ or } 15$$

II). $y^2 - 17y + 70 = 0$

$$y^2 - 10y - 7y + 70 = 0$$

$$y(y - 10) - 7(y - 10) = 0$$

$$(y - 7)(y - 10) = 0$$

$$y = 7 \text{ or } 10$$

So, $x > y$

Hence, the correct answer is option A.

43. Questions

Answer: D

I). $2x^2 + 5x + 3 = 0$

$$2x^2 + 2x + 3x + 3 = 0$$

$$2x(x + 1) + 3(x + 1) = 0$$

$$(2x + 3)(x + 1) = 0$$

$$x = -3/2 \text{ or } -1$$

II). $y^2 + 4y + 4 = 0$

@RankZen Banker

$$y^2 + 2y + 2y + 4 = 0$$

$$y(y + 2) + 2(y + 2) = 0$$

$$(y + 2)(y + 2) = 0$$

$$y = -2 \text{ or } -2$$

So, $x > y$

Hence, the correct answer is option D.

44. Questions

Answer: C

I). $x^2 - 3x - 10 = 0$

$$x^2 - 5x + 2x - 10 = 0$$

$$x(x - 5) + 2(x - 5) = 0$$

$$(x + 2)(x - 5) = 0$$

$$x = -2 \text{ or } 5$$

II). $y^2 + 3y + 2 = 0$

$$y^2 + 2y + y + 2 = 0$$

$$y(y + 2) + 1(y + 2) = 0$$

$$(y + 1)(y + 2) = 0$$

$$y = -1 \text{ or } -2$$

So, the relationship cannot be established.

Hence, the correct answer is option C.

45. Questions

Answer: E

I). $x^2 - 16x + 63 = 0$

$$x^2 - 9x - 7x + 63 = 0$$

$$x(x - 9) - 7(x - 9) = 0$$

$$(x - 7)(x - 9) = 0$$

$$x = 7 \text{ or } 9$$

II). $y^2 - 21y + 108 = 0$

$$y^2 - 12y - 9y + 108 = 0$$

@RankZen Banker

$$y(y - 12) - 9(y - 12) = 0$$

$$(y - 9)(y - 12) = 0$$

$$y = 9 \text{ or } 12$$

$$\text{So, } x \leq y$$

Hence, the correct answer is option E.

@RankZen Banker